

Sunwoo Yu

School of Undergraduate Studies, College of Transdisciplinary studies
Daegu Gyeongbuk Institute of Science and Technology (DGIST)

PERSONAL DATA

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Major	Computer Science		
GPA	3.98/4.3	Major GPA	4.18/4.3

EDUCATION

Daegu Gyeongbuk Institute of Science and Technology (DGIST) **Mar.2020 — Present**

School of Undergraduate Studies, College of Transdisciplinary studies
Bachelor Student

University of California, Los Angeles (UCLA) **Jun.2023 — Aug.2023**

Exchange Program

WORK EXPERIENCE

LG AI Research **Jul.2025 — May.2026**

Bio Intelligence Lab, ML Engineer

✓ Handled large-scale datasets (TB-scale), evaluation & inference pipelines

Neuralscent **Mar.2024 — Dec.2024**

Co-Founder

✓ Built a radar-based sleep dataset and developed sleep stage detection AI model

PUBLICATIONS

* Equal contributor

- Jong Hyun Kim, **Sunwoo Yu**, Juseung Yun, Minsoo Lee, Jongseong Jang, Janghyeon Lee, "Towards a whole-gene bulk transcriptomics foundation model", Nature Communications, *under review*, May. **2026**.
- Minsoo Lee, Jonghyun Kim, Juseung Yun, **Sunwoo Yu**, Jongseong Jang, "MINT: Molecularly Informed Training with Spatial Transcriptomics Supervision for Pathology Foundation Models", Medical Image Computing and Computer Assisted Intervention (MICCAI), *under review*, Mar. **2026**.
- Jong Hyun Kim, **Sunwoo Yu**, Soonyoung Lee, Tae Hyun Hwang, Jongseong Jang, Janghyeon Lee, "Benchmarking gene expression foundation models on bulk RNA-Seq data", Proceedings of the American Association for Cancer Research (AACR), Abstract, Apr. **2026**.
- Juseung Yun, **Sunwoo Yu**, Sumin Ha, Jonghyun Kim, Janghyeon Lee, Jongseong Jang, Soonyoung Lee, [Technical Report] "EXAONE Path 2.5: Pathology Foundation Model with Multi-Omics Alignment", Arxiv, Dec. **2025**.
- Daewon Kim, **Sunwoo Yu**, Jaehyeon Lee and Seonghun Lee, "Development of a Driver Monitoring System Using the Capacitive Steering Wheel and FMCW Radar", The Korean Institute of Communications and Information Sciences (KICS) Conference (Fall), Nov. **2024**.

1. Jaehyeon Lee*, **Sunwoo Yu***, Daewon Kim, and Kiwon Choi, “Multi-patching: life-log Classification with the Reconstructed Representation of Multivariate Time Series”, The 15th International Conference on ICT Convergence (ICTC 2024), pp. 798-803, Oct. **2024**.

RESEARCH EXPERIENCE

- Undergraduate Group Research Program (UGRP)

Feb.2024 — Dec.2024

at Department of Brain Sciences, DGIST
(Prof. Hankyoung Choe, DGIST ABC-DGIST)

✓ Design of a radar dataset collection system for subjects’ sleep in polysomnography experiments

✓ predicting Heart/Respiratory Rate with radar data & predicting sleep stage using representation from HR/RR
- Undergraduate Research Assistant

Sep.2023 — Nov.2024

at Department of Electrical Engineering & Computer Science, DGIST
(Prof. Hoonsung Chwa, DGIST RTCL)

✓ PyTorch Camp : Learning how to use pytorch by implementing below paper
(LaLaRAND: Flexible Layer-by-Layer CPU/GPU Scheduling for Real-Time DNN Tasks)

✓ Project : Real-Time streaming system using foveated rendering

✓ Experience of editing PyTorch & OpenCV source code

AWARDS AND HONORS

- KFAS 人才林 Scholarship

Feb.2025 — Feb.2026
- Encouragement Prize, The World Embedded Software Contest 2024, KESSIA

Nov.2024
- Encouragement Prize, Human Understanding AI Paper Challenge 2024, ETRI

Oct.2024
- Director of ETRI(Electronics and Telecommunications Research Institute) Award
- Academic Excellence Award (DGIST)

Spring Semester, 2024
- Academic Excellence Award (DGIST)

Fall Semester, 2023
- Academic Excellence Award (DGIST)

Spring Semester, 2023

SKILLS

- TEPS:

336/600

valid : Feb.2025 — Feb.2027
- Language:

Python, C/C++
- Tools:

Git, Docker, PyTorch